INCLUSIONS OF THE ZARISKI, ETALE, SMOOTH AND SYNTOMIC COVERING CLASSES IN THE FPPF COVERING CLASS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2135, 2139] and proof(s) [[2141, 2149]]; this spec covers the complete environment instead (statement 2135-2139, proof span(s) [(2141, 2149)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. Definitional inclusion chain with cited morphism facts.

Source reference. Original reference 01UK: sources/stacks-chow/tags.txt, source lines 2413–2413. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01UL: sources/stacks-chow/tags.txt, source lines 2414–2414. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0226: sources/stacks-chow/tags.txt, source lines 2695–2695. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Any syntomic covering is an fppf covering, and a fortiori, any smooth, étale, or Zariski covering is an fppf covering.*

Proof. This is clear from the definitions, the fact that a syntomic morphism is flat and locally of finite presentation, see Morphisms, Lemmas 01UK and 01UL, and Lemma 0226. □

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